

Unsupervised Image-Based Prompting for Lesion Matching Across Mammography Views

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Introduction. During screening mammograms, radiologists will use information from the craniocaudal (CC) and mediolateral oblique (MLO) views to correlate and assess suspicious findings [1]. However, it is quite a challenging task to match findings and has been reported to have an error rate of 23% [2]. With the growing integration of medical software in clinical practice, it becomes important for AI models to integrate such behavior. Existing CAD systems and most AI models still analyse each view independently. Solving the AI matching problem would not only boost performance but also display cross-view matches that assist radiologists and make the AI’s reasoning easier to understand.

To tackle this problem of matching findings, we propose a Vision Transformer (ViT) [3] model trained with some modifications of the DINOv2 [4] framework. Our proposed method provides the following contributions:

1. Extending the DINOv2 [4] framework for processing low-resolution CC and MLO images simultaneously and reinforcing a masked-patch prediction objective that works as a proxy task for matching.
2. A promptable method for using the model’s patch representations to retrieve the most similar region in the corresponding view.

Methods & Experimental Setup. We propose a proxy task where patches are masked in one view (e.g. CC), and lets the model infer their representations from the opposite view (e.g. MLO), encouraging the network to exploit the complementary information. This cross-view task encourages learning spatial correspondences between CC and MLO without requiring matching labels.

Our model was trained on approximately 100,000 paired (CC-MLO) screening mammograms. We provide the DINOv2 backbone with downsampled and concatenated pairs as input and adjusted the IBOT [5] loss function to encourage improved cross-view embedding alignment.

At inference a “source lesion” is a manually delineated (prompted) in one view that we wish to locate in the other. For a given source lesion, we gather all patch tokens within its bounds, compute their cosine similarity against every token in the cross-view image, and then average these similarity heatmaps over the source patches, resulting in a final heatmap that highlights the most likely corresponding region. Figure 1a provides examples of this approach.

Evaluation focuses on malignant and benign cases where both views contain a matching lesion. Using previously available ground-truth (GTR) segmentations

rather than the inference-time manual prompts, we report the percentage of correct cross-view links whose predicted center falls inside the true lesion footprint (Figure 1b).

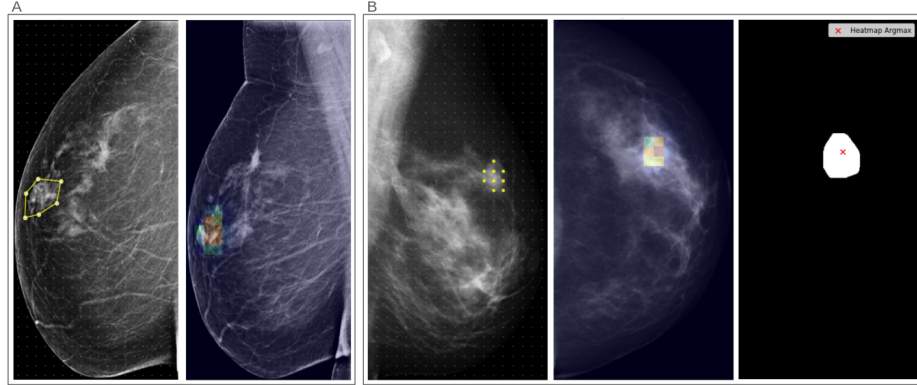


Fig. 1. a) Inference: CC view with a manually prompted lesion contour and the resulting heatmap over the MLO view. . b) Evaluation linking: using existing GTRs in the MLO view with CC heatmap and peak prediction in GTR.

Results. On a held-out set of 938 paired lesions (malignant and benign), we achieve 78.0 % correct links (center of predicted region within the ground-truth footprint; 95 % CI: 75.3 – 80.6 %). Our unsupervised approach demonstrates results which are comparable to the error-rate of radiologists report in [2].

Conclusion. Our method offers a novel approach to address the challenging task of lesion correspondence in a purely unsupervised manner. Leveraging a pretrained foundational model, we demonstrate that our model learns a shared embedding between corresponding lesions across CC and MLO views, and reported 78.0% correct linking of candidate findings. Such a method may generalize to improve cancer classification or to other tasks such as link findings across modalities.

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